

ChE 494 (Entrepreneurship in Chemical Engineering)

Department of Chemical Engineering

University of Illinois at Chicago

Fall 2019

Instructors:

Said Al-Hallaj, PhD

Course Time: Thursday 3:30 to 6:15 pm

Office Hours: By appointment

Prerequisite: course suitable for senior and graduate engineering students.

Credit Hours: 4 hr (2-credit for lecture and 2-credit for case study)

Course Description:

The course provides an introductory view into the various aspects involved in technology commercialization (i.e. taking an idea from the lab to the marketplace). Following is an outline of the course outline:

Part I: Intellectual Property

- 1) Confidentiality of Non-Disclosure Agreement (NDA)
- 2) Patent applications
- 3) Copyrights
- 4) Know-how

Part II: UIC Resources

- 1) Office of Technology Management
- 2) Invention Center
- 3) Business Plan Competition
- 4) Illinois Venture

Part III: Business Development Strategy

- 1) Business Plan
- 2) Marketing strategies
- 3) Finance and accounting

Part IV: Starting a Business

- 1) Legal structure (LLC, Inc., etc)
- 2) Operating Agreement
- 3) Governance

Part V: Fundraising

- 1) Grants
- 2) Seed/Angel investment
- 3) Venture capital
- 4) Strategic partnerships

Text Books: TBD

Grade:

25% Weekly assignments (3-4 mini-projects or written essays)

25% Midterm Home Exam

50% Final Project: written plan (25%) and oral presentation (25%)

Final Project: technology commercialization strategy (i.e. IP assessment, market study, go-to-market strategy)

Class Schedule

<u>Week</u>	<u>Topic</u>
8/29	Introduction (Anderson Entrepreneurship Quiz)
9/5	Protecting your intellectual property (IP) (Guest Lecturer: Mark Krivchenia, UIC-OTM)
9/12	Case Studies Presentations (<i>form case study teams</i>)
9/19	UIC Resources Overview (Guest: Mark Krivchenia, UIC-OTM) (<i><u>Final Project Abstract Due</u></i>)
9/26	Business Development Strategy (Mike Pintar, COO NETenergy)
10/3	Why Marketing Study is Important
10/10	Financials Basics Steve Wolff, CFO AllCell Technologies, LLC
10/17	Starting Your Own Business (is it a good idea?)
10/24	How to build an effective team
10/31	Midterm (<i><u>Final Project Pitch</u></i>)
11/7	Corporate Entrepreneurship
11/14	Pitching your idea
11/21	Fundraising Strategy (are you ready to pitch your idea?)
11/28	Happy Thanksgiving!
12/5	Final (Oral Presentations)
12/12	Finals Week (written report due)